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# CHEMISTRY

TARGET : JEE-ADVANCED-2024

# CAPS-8+

**SCQ (Single Correct Type) :**

**1.** The reaction A(g) + 2B(g)$\rightarrow$C(g) is an elementary reaction. In an experiment involving this reaction, the initial partial pressures of A and B are $P_A$ = 0.40 atm and $P_B$ =1.0 atm respectively. When pressure of C becomes ($P_C$) = 0.3 atm in the reaction the rate of the reaction relative to the initial rate is

(A) $\frac{1}{12}$ (B) $\frac{1}{50}$ (C) $\frac{1}{25}$ (D) none of these

**2.** What is the % decrease in vapor pressure with respect to solution when saturated solution of $Ag_3PO_4$ is prepared

Given : $K_{sp}$ of $Ag_3PO_4 = \left(\frac{3}{16}\right)^3 \times 10^{-12}$ (Assume no hydrolysis)

(A) $6 \times 10^{-4}$×× (B) $1.5\times10^{-4}$ (C) $9\times10^{-4}$ (D) $18\times10^{-4}$

**3.** Consider the 1st order reactions :

$A\xrightarrow[E_1]{k_1}B$ ; $2A\xrightarrow[E_4]{k_4}D$ ; $B\xrightarrow[E_2]{k_2}C$ ; $D\xrightarrow[E_3]{k_3}2C$

$E_1$, $E_2$, $E_3$ and $E_4$ are respective activation energies and $k_1$, $k_2$, $k_3$ and $k_4$ are respective rate constants. Which of the following is/are correct ?

(A) $\frac{d[B]}{dt} = K_1[A] - k_2[B]$×× (B) $\frac{d[C]}{dt} = K_2[B] + 2k_2[D]$

(C) Average activation energy of $A = \frac{k_1E_1 + k_4E_4}{k_1 + k_4}$

(D) None of these

**4.** Consider the following metallurgical processes

(i) Heating impure metal with CO and distilling the resulting volatile carbonyl (boiling point 43ºC) and finally decomposing at 150ºC to 200ºC to get the pure metal.

(ii) Heating the sulphide ore in air until a part is converted to oxide and then further heating in the absence of air to let the oxide react with unchanged sulphide.

(iii) Electrolysing the molten electrolyte containing approximately equal amounts of the metal chloride and $CaCl_2$ to obtain the metal.

The process used for obtaining sodium, nickel and copper are, respectively,

**5.** $H_2S$ and $SO_2$ can be distinguished by :

(A) Litmus paper (B) $MnO_4^-$ (C) $(CH_3COO)_2Pb$ (D) All of these

**MCQ (Multiple Correct Type) :**

**6.** A substance ‘A’ has the following variation of vapor pressure with temperature for its solid state and liquid state. Identify the options which are correct :

Given for solid A : $\log_{10} P = 4 - \frac{2000}{T}$ ; For liquid A : $\log_{10} P = 3.5 - \frac{1500}{T}$ where P is in mm of Hg and T in K

(A) Triple point temperature of substance 'A' is 1000 K

(B) $\Delta H_{sub}$ will be approximately 9.212 Kcal/mol

(C) $\Delta H_{fusion}$ will be approximately 9.212 Kcal/mol

(D) $\Delta H_{vap}$ will be approximately 6.909 Kcal/mol

**7.** An ideal binary liquid solution of A and B has a vapour pressure of 900 mm of Hg. It is distilled at constant temperature till $\frac{2}{3}$ of the original amount is distilled. If the mole fraction of ‘A’ in the residue and mole fraction of B in the condensate is 0.3 and 0.4 respectively and vapour pressure of residue is 860 mm of Hg, then which of the following is/are correct options :

(A) The initial mixture taken should be an equimolar mixture of A and B.

(B) The first vapors formed will have more moles of A as compared to moles of B.

(C) $P_{A^{\circ}}^{\circ}$ = 1000 torr

(D) The vapor pressure above the condensate is 920 torr at same temperature.

**8.** Which of the following is/are correctly matched with respect to the processes involved in the extractive metallurgy of the respective metal ?

(A) Bauxite $\rightarrow$ Al : Leaching, precipitation, calcination and electrolytic reduction (molten state).

(B) $Ag_2S \rightarrow$ Ag : Leaching and displacement method.

(C) PbS $\rightarrow$ Pb : Froth flotation process, roasting and self reduction.

(D) $KCl.MgCl_2.6H_2O \rightarrow$ Mg : Dehydration by simple heating and electrolytic reduction (molten state).

**9.** Suppose $Cu^{2+}$ and $Pb^{2+}$ both are present in a mixture. Then they can be separated by :

(A) adding KI solution (B) adding $NH_3$ solution

(C) adding NaOH solution (D) adding dilute HCl (2M)

**10.** Heating which of the following salts in a dry test tube may cause a change in their colour ?

(A) $ZnCO_3$ (white) (B) $Co(NO_3)_2.6H_2O$ (red)

(C) $FeSO_4.7H_2O$ (green) (D) $MnSO_4$ (faint pink)

**11.** Which of the following is/are false?

(A) In Gold Schmidt's thermite process aluminium acts as a reducing agent.

(B) Mg is extracted by electrolysis of aqueous solution of $MgCl_2$.

(C) Extraction of Pb is possible by smelting of its oxide.

(D) Red Bauxite is purified by Serpeck’s process.

**Paragraph (One or more than one correct options) Que. 12 to 13**

A pungent smelling gas ‘X’ is produced when a salt ‘P’ is treated with concentrated $H_2SO_4$. The gas ‘X’ is colorless and also give dense white fumes with $NH_3$. The solution of salt P gives white precipitate with $AgNO_3$.

The white precipitate dissolves in dilute $NH_3$. Gas X gets oxidised by oxygen in the presence of $CuCl_2$ to produce gas 'Y' and liquid 'Z' at room temperature.

**12.** Which of the following is/are correct about gas X ?

(A) X react readily with sodium carbonate. (B) X is an oxidising agent

(C) X produces acidic solution in water (D) X is not oxidized by ferric chloride.

**13.** Gas Y reacts with hypo solution to produce gas X and species W Hence.

(A) $Y = Cl_2$ (B) $W = Na_2SO_3$ (C) $W = NaHSO_4$ (D) $Y = SO_2$

**Paragraph (One or more than one correct options) Que. 14 to 15**

The mineral colemanite is fused with sodium carbonate, compound (Z) is obtained along with white precipitate. When (Z) reacts with dil. $H_2SO_4$ gives a compound (A) which on strong heating gives an oxide (C). (C) on reduction with Mg produced (D) and non metal (X). Treatment of chlorine on a mixture of (C) and carbon at high temperature gives a halide (E) which is fuming liquid (boiling point 13ºC) along with a gas (F). (E) is a Lewis acid.

**14.** (Z) may be :

(A) $H_3BO_3$ (B) $BaCO_3$ (C) borax (D) $Na_3BO_3$

**15.** (A) and (C) may be :

(A) $B_2H_6$,B (B) $B_2H_6$, $B_2O_3$ (C) $H_3BO_3$,B (D) $H_3BO_3$, $B_2O_3$

## Match the column

**16.** Match the compound in column (I) with the properties of products obtained on heating it in column (II).

| | Column - I | | Column - II |
|---|---|---|---|
| (A) | $FeSO_4$ | (p) | One of the product is basic oxide |
| (B) | $NH_4NO_3$ | (q) | One of the product is acidic oxide |
| (C) | $Ba(N_3)_2$ | (r) | One of the product is unreactive towards $O_2$ at room temperature. |
| (D) | $MgCl_2.6H_2O$ | (s) | At least one of the product is neutral oxide. |

**(Numerical Value type ) :**

**17.** The number of compounds/elements oxidised by $XeF_2$ among following is :

HF, HBr, HCl, HI, $NH_3$, $CrF_2$, Pt, $S_8$

**18.** How many of the following acids may undergo disproportionation reaction on heating ?

$H_2C_2O_4$ , $H_3PO_2$ , $H_3PO_3$ , HClO , $HNO_2$ , $H_2SO_3$ , $H_2SO_4$ , $HClO_3$

**19.** $SOCl_2$ can react with how many of the following species to liberate $SO_2$?

$H_2O$, HCl, $C_2H_5OH$, HBr, $CH_3COOH$, HCN, $H_2SO_4$, $H_3PO_4$, $D_2O$, HI, HF

**20.** H-F is a weak acid but on addition of $AsF_5$ , it becomes a very strong acid. The number of 90ºº angles in the anionic part of the product is_________.

**21.** $NaOH + PbO \xrightarrow{\Delta} X + H_2O$

$NaOH + SnO_2 \xrightarrow{\Delta} Y + H_2O$

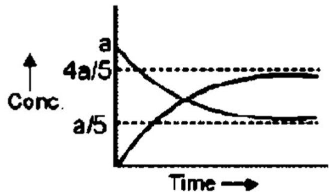
$NaOH + H_2O + Al \xrightarrow{\Delta} Z + H_2$

Sum of the number of atoms present in one molecule each of X, Y, Z is..........

**22.** Amongst the following the total number of ions which produce colour in the solutions (in water) is.

$_{22}Tl^{3+}$, $_{23}Cr^{4+}$, $_{24}Cr^{3+}$, $_{82}Pb^{2+}$, $_{26}Fe^{2+}$, $_{30}Zn^{2+}$, $_{28}Ni^{2+}$, $_{21}Sc^{3+}$, $_{80}Hg^{2+}$,

**23.** A reversible reaction which is I$^{st}$ order in both direction $A \underset{K_2}{\overset{K_1}{\rightleftharpoons}} B$. For which value of $\frac{K_1}{K_2}$ following graph will be correct (where a is initial amount of reactant A)




**24.** Amongst the following, the total number of compounds whose aqueous solution gives white precipitate with $Pb^{2+}$ ions is :

KI, $NH_3$, $Na_2CO_3$, $K_2CrO_4$, NaCl, $Na_2SO_3$, $Na_2S$, $KNO_3$, $NaClO_4$

**25.** How many of the following metallurgies involve leaching ?

Bauxite $\longrightarrow$ Al ; $Ag_2S \longrightarrow$ Ag ; Imp. Au $\longrightarrow$ Au ; $CuFeS_2 \longrightarrow$ Cu ; PbS $\longrightarrow$ Pb

$MgCl_2 \longrightarrow$ Mg ; $FeCO_3 \longrightarrow$ Fe ; Low grade copper ore $\longrightarrow$ Cu ; HgS $\longrightarrow$ Hg

## Answer Key

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | (C) | 2. | (C) | 3. | (A) | 4. | (C) | 5. | (B, C) |
| 6. | (ABD) | 7. | (ABCD) | 8. | (ABC) | 9. | (ABCD) | 10. | (ABCD) |
| 11. | (BD) | 12. | (ACD) | 13. | (AC) | 14. | (C) | 15. | (D) |
| 16. | (A) $\rightarrow$ p, q, r ; (B) $\rightarrow$ r, s ; (C) $\rightarrow$ r ; (D) $\rightarrow$ p, r, s | | | | | | | 17. | 7 |
| 18. | 6 | 19. | 6 | 20. | 12 | 21. | 15 | 22. | 5 |
| 23. | 4 | 24. | (4) | 25. | 4 | | | | |